Venkata Lakshmi Parimala Pasupuleti

Full Stack Software Engineer · Affordable Housing Domain Expert · HUD · LIHTC · Compliance Systems · Regulated SaaS · React · Python · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Fairfax, Virginia, USA

SUMMARY

Full-stack software engineer with 9+ years of experience, including **7 years building compliance-critical software in affordable housing SaaS** - the exact domain AppFolio serves. At Entrata (100K+ users, regulated multi-tenant affordable and market-rate housing platform), owned HUD income-limit compliance ETL pipelines (180K records, zero violations across 6 years), built LIHTC certification tracking and compliance dashboards used by property managers across large portfolios, and engineered PHI-compliant data systems under HUD regulatory frameworks. Deep familiarity with the complexity of affordable housing compliance: recurring certification workflows, regulatory submission requirements, and ever-shifting income limits across jurisdictions. Currently at Intuit integrating AI into production financial services. Strong full-stack: Python, PHP/Laravel, Java backend + React/TypeScript frontend; PostgreSQL/MySQL, AWS, Docker, Kubernetes, microservices (SOA). Practiced in TDD, code reviews, and Agile delivery across 120+ sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Affordable Housing Domain · HUD Compliance · LIHTC · Regulatory Workflows · Certification & Recertification Tracking · Income Limit Compliance · PHI

Regulated Multi-Tenant SaaS · Compliance Systems · Property Management · Full-Stack Web Applications · SaaS Engineering · End-to-End Ownership

React · TypeScript · Frontend Development · Component Architecture · Python · PHP/Laravel · Java · Spring Boot · Backend Engineering

PostgreSQL · MySQL · SQL Optimization · Database Design · AWS (S3 · SQS · Lambda · EC2) · Docker · Kubernetes · Microservices (SOA)

Test-Driven Development · Code Quality · CI/CD · GitHub Actions · Agile · Scrum · Kanban · Code Reviews · Mentorship

Data Pipelines · ETL · Async Processing · RabbitMQ · Event-Driven · AI Integration · LLM Workflows · Evaluation Infrastructure

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- Integrate Anthropic Claude into production financial services - autonomous AI workflows, tool calling, retrieval-grounded context management, structured outputs, guardrails; LLM-as-judge evaluation harnesses with CI regression gating.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; Docker + GitHub Actions CI/CD to AWS with structured logging and CloudWatch monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing & Apartment SaaS (regulated multi-tenant, 100K+ monthly active users)

- **HUD compliance & affordable housing domain:** Owned HUD income-limit compliance ETL pipeline for 6+ years (180K records across jurisdictions, zero compliance violations); built LIHTC certification tracking dashboards for property managers - certification status, eligibility checks, and regulatory submission workflows; engineered PHI-compliant AWS S3 data migration (Libsodium encryption, 100% coverage, zero incidents) under HUD regulatory requirements.
- **Full-stack SaaS engineering:** Led Docker microservices redesign (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries across PostgreSQL schemas (30% performance gain); RabbitMQ async event-driven processing for rent updates across thousands of tenant records - retry logic, idempotency guarantees, dead-letter routing.
- Python/Bash automation eliminating 80% of manual compliance workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints; owned features from design through production.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL) for affordable housing compliance features; delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Agentic Compliance Platform: LangGraph · RAG · HITL · Compliance Workflows 2026 | Python, LangGraph, FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic platform for lease compliance workflows: 8-agent LangGraph orchestration with tool calling, durable state, and conditional routing; hybrid search RAG (OCR ingestion, TF-IDF + semantic reranking) over complex compliance documents; HITL guardrails enforcing human oversight on regulated decisions; CI eval harness (LLM-as-judge, golden datasets: 93.6% precision, 1,240 test cases). Built on deep familiarity with housing compliance complexity.

AgentProbe - LLM Evaluation & Observability Platform 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source eval and observability platform: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, failure handling; cost-aware multi-provider routing (Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, Ollama); full per-run tracing and structured logging.

Real Estate Data Analysis — ML on Property Price & Socioeconomic Factors 2024 | Python, XGBoost, Random Forest, scikit-learn, Pandas | George Mason University

- XGBoost + Random Forest models on Connecticut property transaction data (2001-2022) analyzing socioeconomic impact on housing prices. Feature engineering across 15+ variables; model evaluation with cross-validation; findings presented to faculty. Directly relevant to housing market dynamics and affordability analysis.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017